OCR-Based Scoring Conceals Rendering Failures in Text-to-Image Evaluation: Evidence from Hangul

A measurement-validity study

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Abstract

Text-to-image (T2I) benchmarks for visual text rendering routinely score generated images with an OCR engine. We show that this practice has a structural defect: recognition-based judges are closed-set classifiers that must return *some* character from a known inventory, and therefore cannot report the single most informative failure mode — that the model drew a glyph that **does not exist in the writing system at all**.

Using 300 Hangul syllable images from a commercial T2I model, exhaustively labelled by two human annotators (binary validity: Krippendorff's $\alpha = 0.942$), we find that **21.7% of outputs are not valid Hangul syllables** — they are Hangul-shaped glyphs that cannot be typed, e.g. a final consonant ㅁ drawn with an internal cross. Three OCR engines with independent architectures (Naver CLOVA, EasyOCR, PaddleOCR) each map 77–83% of these non-existent glyphs onto a valid syllable, and in 77% of jointly-answered cases the two engines snap to *different* characters — indicating the cause is the closed-set formulation itself rather than a shared dictionary.

The consequences for benchmark numbers are large. Relative to human ground truth, OCR-only scoring understates absolute accuracy by 25.6–50.0 percentage points, and because the understatement grows with orthographic complexity, it **exaggerates the complex-coda penalty by 2.3×–4.0×**. Effect-size estimates also depend on engine choice, so results from papers using different OCR engines are not comparable.

To isolate the judge property from any particular generator, we run a controlled paired experiment using no generative model at all: clean font renderings (control) versus the same glyphs with strokes programmatically added or removed (treatment, invalid by construction). EasyOCR's answer rate is statistically indistinguishable between conditions (96.4% vs 93.0%; $p = 0.234$; equivalence test at ±10pp margin passes), and its confidence distribution is likewise indistinguishable (0.458 vs 0.464, Mann–Whitney $p = 0.841$). The engine cannot tell a real character from a fabricated one, and its confidence carries no signal about the difference.

The implication is not that a more accurate OCR is needed: concealment is decoupled from accuracy (the most accurate of the three engines still conceals 78.5%), because the defect lies in the closed-set output formulation rather than in recognition quality. What is needed instead is an evaluation protocol built around the limitation — reporting "not a valid character" as a distinct outcome, and quantifying rather than hiding the human fraction it requires.

We release the human-labelled gold set, all code, and a verification script that recomputes every number in this note from raw data.

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1. Problem

Benchmarks that measure how well image generators render text (AnyText, Qwen-Image's evaluation suite, and Hangul-specific efforts) share a common pipeline: prompt the model with a target string, run OCR on the output, compare the OCR reading to the target. The OCR engine is treated as a transparent measuring instrument.

It is not. An OCR engine trained to recognise a script has a fixed output inventory — for Hangul, the 11,172 precomposed syllables of the Unicode Hangul Syllables block. Confronted with an image containing a glyph outside that inventory, the engine has no way to say "none of the above". It returns its nearest neighbour.

This matters because generative models fail in exactly that way. They do not merely draw the *wrong* character; they draw characters that **do not exist**. One annotated case: asked for '감' (gam), the model produced a glyph whose final consonant ㅁ contained an internal cross, making it '田' — not a Hangul jamo. The human annotator classified this as "not typeable". CLOVA and a shape-matching baseline both read it as '감' and scored it correct.

Figure 1 shows this pair as generated. The vendor and model identifier are withheld pending written clarification on redistribution terms (§9); both images were produced by the same generator under the same prompt template, target syllable '감', with only the sampling draw differing.

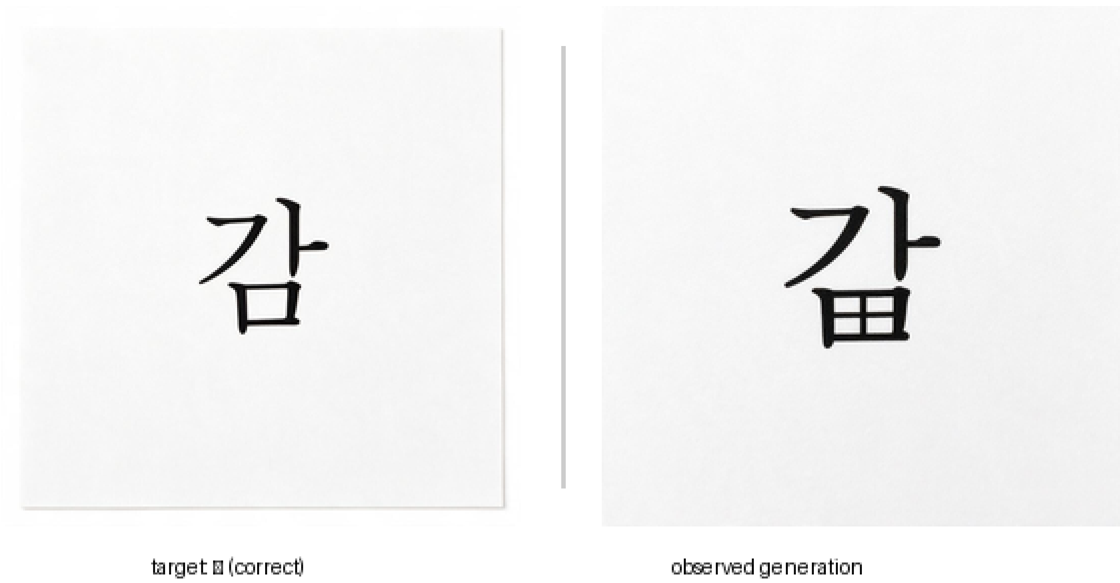

Figure 1. Left: a correctly formed output for the same target. Right: the observed failure — an invalid glyph resembling '田' in place of the coda. Generating model withheld pending provider clarification (§9); the controlled reproduction of this failure type used for the causal experiment in §5 uses only OFL font renderings (Figure 2) and does not depend on this image.

If the dominant failure mode is invisible to the instrument, the benchmark measures something other than what it claims to.

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2. Data

Generator. A commercial text-to-image model accessed via a paid API, 300 images, two prompt templates, single Hangul syllable targets drawn from an 18-cell stratified design (2 vowel classes × 3 coda classes × 3 onset groups; see `docs/partitioning.md`). We withhold the vendor and model identifier here pending written clarification from the provider on redistribution and publication terms (see `docs/RELEASE.md`); this is a disclosure-scope choice, not an attempt to obscure the finding, and no per-vendor claim is made — §7 states explicitly that this note does not rank or characterise any specific model.

Human ground truth. All 300 images were labelled target-blind. The primary criterion, fixed after a pilot revealed that "illegible" was being used inconsistently, is:

> ***Can this glyph be typed on a standard keyboard?*** *— i.e. is it a member of the 11,172 valid precomposed syllables?*

This is a binary, reproducible criterion, not a subjective legibility judgment. Annotators transcribed the character only when they judged it valid.

Reliability. Two annotators independently labelled an 87-item subset.

Judgment	Agreement
Binary validity (valid / not)	97.7% raw; **Krippendorff's α = 0.942**
Transcription, given both said valid	**23/23 = 100%**
4-way subcategory (malformed / non-Hangul / multi / none)	α = 0.576

The binary axis is highly reliable and the transcription that follows it is perfect. The finer subcategories are **not** reliable — the boundary between "malformed Hangul" and "not Hangul" is a continuum, and 17 of 21 disagreements fall on exactly that boundary. We therefore report only the binary distinction as a primary measure and treat subcategories as exploratory.

Label distribution (n = 300).

Label	n	%
valid syllable	235	78.3%
malformed (Hangul-structured, invalid strokes)	42	14.0%
non-Hangul (symbols, shapes)	13	4.3%
multiple syllables	10	3.3%
not a valid syllable (total)	**65**	**21.7%**

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3. Result 1 — Three engines conceal invalid glyphs at the same rate

For each of the 65 human-invalid images we ask: does the engine nonetheless return a single valid Hangul syllable?

Engine	Architecture	Concealment rate	False positives (matched target)
CLOVA General OCR	commercial, closed	50/65 = **76.9%**	2
EasyOCR	CRNN, open source	54/65 = **83.1%**	3
PaddleOCR (`korean_PP-OCRv5`)	PP-OCR, open source	51/65 = **78.5%**	3

The three engines share neither training data nor architecture, yet converge on 77–83%.

The cause is the closed-set formulation, not a shared lexicon. Of the 44 invalid images where both CLOVA and EasyOCR returned an answer, **34 (77.3%) returned *different* characters**. They do not agree on what the glyph is; they only agree that it must be *something*.

Examples (target → CLOVA / EasyOCR): 퐋 → 쨰 / 팩 · 쥢 → 쥣 / 젓 · 쬐 → 쩍 / 짜.

Figure 3. Target '퐋' = ㅍ + ㅘ (a two-stroke compound vowel, ㅗ+ㅏ) + ㄳ (a two-consonant coda cluster). What was actually drawn is onset ㅍ followed by two separate vowel strokes — one shaped like ㅗ, the other showing two tick marks like ㅑ — a pairing that is not one of Hangul's seven legal compound vowels (ㅘㅙㅚㅝㅞㅟㅢ); ㅗ + ㅑ does not exist as a diphthong. The coda is also simplified, from the target's cluster ㄳ to a plain ㄹ. Because no valid syllable contains both drawn vowel strokes, CLOVA and EasyOCR each keep only one: CLOVA reads '폴' (ㅍ+ㅗ+ㄹ), matching the ㅗ-shaped stroke; EasyOCR reads '팔' (ㅍ+ㅏ+ㄹ), the closest single vowel to the ㅑ-shaped stroke. The two engines are not disagreeing over one ambiguous vowel — each discards a different real stroke from a glyph whose vowel component has no valid single-character reading at all.

Note that false positives — cases where the concealed reading happens to match the target and thus inflates the score directly — are rare (2–3 of 65). Concealment damages measurement mainly by *erasing a failure category*, not by awarding spurious credit.

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4. Result 2 — Understatement grows with complexity, exaggerating structural effects

Exact-match accuracy against the target, unconditional (invalid glyphs count as wrong), n = 300:

Coda class	Human	CLOVA	EasyOCR	PaddleOCR
no coda	70.0%	44.4%	43.3%	61.1%
simple coda	60.8%	30.0%	17.5%	27.5%
complex coda	55.6%	8.9%	5.6%	10.0%
overall	**62.0%**	28.0%	21.7%	32.3%

Understatement (human − engine), paired bootstrap, 95% CI:

	no coda	simple coda	complex coda
CLOVA	+25.6 [15.6, 35.6]	+30.8 [22.5, 39.2]	**+46.7 [36.7, 57.8]**
EasyOCR	+26.7 [16.7, 36.7]	+43.3 [34.2, 52.5]	**+50.0 [38.9, 61.1]**
PaddleOCR	+8.9 [1.1, 17.8]	+33.3 [25.0, 41.7]	**+45.6 [34.4, 56.7]**

Every interval excludes zero. Critically, **the understatement is not constant** — it grows with orthographic complexity. Because a structural comparison is a difference of two accuracies, a complexity-correlated bias does not cancel:

	complex − simple coda gap	Exaggeration	Ratio
Human (truth)	**−5.3pp**	—	1.0×
CLOVA	−21.1pp	−15.8pp, CI [−29.4, −1.4] **significant**	**4.0×**
PaddleOCR	−17.5pp	−12.2pp, CI [−25.3, +1.7] n.s.	3.3×
EasyOCR	−11.9pp	−6.7pp, CI [−21.1, +7.8] n.s.	2.3×

All three engines overstate the complex-coda penalty; for CLOVA the exaggeration is statistically significant at n = 300. **The exaggeration factor differs by engine (2.3×–4.0×), so effect sizes reported by studies using different OCR engines are not comparable.**

This is the practically important claim of this note. A paper concluding "complex orthography degrades rendering by X%" using OCR scoring may be reporting a number several times the true effect, with the multiplier set by an arbitrary tooling choice.

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5. Result 3 — Controlled experiment: engines cannot detect fabricated glyphs

Results 1–2 are observational: they compare judges against human labels on one generator's output. To establish the judge property causally and independently of any generator, we run a paired experiment with **no generative model**.

Design.

Control	clean font rendering of a valid syllable (n = 84)

Treatment	*the same syllable*, with strokes programmatically added or removed at three severities (n = 497)
Randomisation	syllable, perturbation site, and severity all seeded RNG
Controlled	font, resolution, glyph scale, ink volume — same base glyph
Ground truth	invalid **by construction**; a template-match post-check discards samples that accidentally reconstruct a different real syllable (7 rejected)

Perturbations mirror failure types recorded in the human audit notes ("a stroke of the final ㅂ is missing", "unknown strokes added to ㅡ").

Figure 2. The same syllable ('값') under all five conditions of the paired design, generated by `synthetic_malformed.make_item()` *and rendered with identical print-like degradation (same seed) across all five panels so the comparison isolates the perturbation, not incidental rendering noise. Only the leftmost image is a real syllable; the other four are invalid by construction. The severity gradient from mild to severe is visible in stroke count and displacement.*

Findings (EasyOCR).

	Control	Treatment
Returns a valid syllable	96.4%	93.0%
Mean confidence	0.458	0.464

- Answer rate: difference −3.5pp, z = −1.19, **p = 0.234**. Because the hypothesis of interest is *absence* of discrimination, we additionally run a TOST equivalence test at a ±10pp margin: the 95% CI [−9.2, +2.2] lies entirely within the margin — **equivalence is positively established**, not merely "failed to reject".
- Confidence: Mann–Whitney **p = 0.841**. The confidence distribution when looking at a real character is statistically identical to when looking at a fabricated one. **Confidence carries no signal for this distinction** — ruling out the obvious mitigation of thresholding on confidence.

Findings (template matching, closed-set shape baseline). Control accuracy 96.4%; on perturbed glyphs it snaps back to the unperturbed original 94.0% of the time. A purely geometric matcher with no language model shows the same behaviour, confirming the mechanism is closed-set forced choice rather than linguistic priors.

Sensitivity. The engines are not wholly insensitive: for stroke removal, snap rate falls from 98.8% (mild) to 81.0% (severe), p = 0.0002. Detection is a matter of degree. But the near-miss regime — one stroke wrong — is both where detection fails and where real generator failures live.

A third engine (PaddleOCR) was run on a reduced replication (n = 126) and reproduces the direction (control 88.9% vs treatment 84.3%, p = 0.611; confidence 0.669 vs 0.591, p = 0.351) but with a control group too small (n = 18) to pass the strict equivalence test. We report EasyOCR (n = 84 control) as the confirmatory result and PaddleOCR as directional replication.

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6. A partial mitigation, and its limits

We tested whether a confidence gate can salvage automatic scoring: accept the OCR reading when confidence ≥ 0.80, route everything else to a human. Two refinements were required, both discovered empirically:

- The gate is **self-correcting on the coda axis** — CLOVA loses confidence precisely where it is unreliable, so hard cases route to humans and bias shrinks rather than grows.
- It **fails for tensed onsets** (ㄲㄸㅃㅆㅉ), where CLOVA is *confidently* wrong: disagreement with humans in the auto-accepted region is 47.4%, versus 4.5% for plain onsets. Excluding tensed onsets from auto-acceptance brings bias spread on all three cell axes below 2.5pp.

Resulting protocol: **50% of images still require human judgment.** We report this not as a solution but as a measured lower bound on automation for this task. Crucially, the protocol still cannot detect invalid glyphs: of 128 auto-accepted images, 16 (12.5%) are human-invalid, inflating the reported valid-syllable rate from a true 78.3% to 83.7%. **The concealment effect reproduces inside our own result table**, which is why every automatically scored figure must be published with its measured bias.

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7. What this note does and does not claim

Claims.

1. Recognition-based judges conceal non-existent glyphs at 77–83%, replicated across three independent architectures (§3).
2. OCR-only scoring understates accuracy by 25.6–50.0pp, with the understatement growing with complexity, exaggerating structural effects 2.3×–4.0× (§4).
3. Engines cannot distinguish fabricated glyphs from real ones, and their confidence carries no signal for the distinction — established causally in a controlled, generator-free paired experiment (§5).
4. Human binary validity judgment is reliable (α = 0.942) and is also the *cheapest* human judgment, since it requires no transcription.

Explicitly not claimed.

- **Not a model benchmark.** One generator, 300 images. We make no claim about this or any model's ranking. The generator is an instance, not the subject.
- **No causal claim about orthographic structure.** The finding that complex codas are harder is observational and confounded with character frequency and stroke count. We could not separate these, and a prior version of this project mistakenly attributed an OCR scale artefact to coda complexity before the confound was found.
- **Per-cell (18-cell) values are not reported.** At ~12 images per cell, confidence intervals are uninformative. Only the three marginal axes are reported.
- **The cross-script generalisation in §8.2 is a prediction, not a result.** We tested Hangul only. No Chinese, Japanese, Devanagari or Arabic data were collected; §8.2 states a mechanism-based expectation and supplies a method to test it.
- **The decoupling claim in §8.1 rests on three engines.** We claim only that concealment does not track accuracy, not that they are inversely related.
- **The tensed-onset exclusion rule is in-sample.** Derived from n = 63 and evaluated on the same data; it needs out-of-sample confirmation.
- **One primary annotator is the author.** The second annotator is independent; agreement statistics are computed between them. This is a limitation, disclosed.

8. Implications

8.1 The fix is not "a more accurate OCR"

The natural reading of these results is that we need a better OCR engine. The data argue otherwise: **concealment is decoupled from accuracy.**

Engine	Accuracy (valid glyphs)	Concealment (invalid glyphs)
PaddleOCR	**32.3%** (best)	78.5%
CLOVA	28.0%	**76.9%** (lowest)
EasyOCR	**21.7%** (worst)	**83.1%** (highest)

Accuracy varies by a factor of 1.49 across the three engines; concealment varies by only 1.08. The *most* accurate engine still conceals 78.5% of non-existent glyphs. (With n = 3 engines we do not claim a monotonic relationship — the rank orders are not simply inverted — only that concealment does not track accuracy.)

This is what one would expect from the mechanism. Concealment is not a recognition error to be trained away; it is a consequence of the **output formulation**. A classifier whose label space is the 11,172 valid syllables has no symbol for "none of these", so the question "is this a character at all?" is not merely answered badly — it is not asked. §5 confirms this on the judge's own confidence signal: the confidence distribution on fabricated glyphs is statistically identical to that on real ones ($p = 0.841$), so no threshold on any post-hoc confidence score can recover the distinction.

Making the label space open — open-set recognition, or a calibrated reject option — is a live and unsolved problem in machine learning generally, not a matter of more training data for Korean. **Treating "get a better OCR" as the action item defers the problem indefinitely.** The tractable action is to design the *evaluation protocol* around the limitation, which is what §6 and §8.3 describe.

8.2 Beyond Hangul: a prediction for other compositional scripts

Hangul is a convenient test case because its composition rule is explicit and its valid inventory is exactly enumerable ($19 \times 21 \times 28 = 11{,}172$), which is what makes "this glyph is not in the inventory" a decidable, reproducible label. But nothing in the mechanism is Hangul-specific.

We therefore **predict, without having tested it**, that the same concealment occurs for any writing system where characters are assembled from reusable sub-parts and the recogniser's label space is the set of assembled characters — Chinese radical composition, Japanese kanji, Devanagari and other Brahmic conjuncts, Arabic ligature forms. A generator that draws a plausible-but-nonexistent composition in those scripts should be silently mapped onto its nearest real character, for exactly the reason it is here.

This matters because the major visual-text-rendering benchmarks (AnyText, the Qwen-Image evaluation suite, and their successors) score Chinese with OCR pipelines of the same architecture family as the ones tested here.

The prediction is cheap to test, and we release the means to do it: the `synthetic_malformed` procedure (§5) requires only a font and a list of valid characters, uses no generative model, and yields ground truth by construction. Porting it to another script is a matter of substituting the character inventory and the perturbation primitives.

8.3 Recommendations

For anyone building or reading a visual-text-rendering benchmark:

1. **Validate the judge before scoring the model.** Compare the judge to human labels on a sample from the same distribution. This project made three separate wrong attributions — including attributing an OCR font-scale artefact to model weakness — before the pipeline was checked.
2. **Test whether the judge can abstain**, not just whether it is accurate. Accuracy on valid characters says nothing about behaviour on invalid ones, and the latter is where generators fail. §8.1: picking the most accurate engine does not help.
3. **Report "not a valid character" as a first-class outcome**, separate from "wrong character". They are different failures with different causes, and the distinction costs nothing to add to a results table.
4. **Do not compare effect sizes across studies using different OCR engines.** The exaggeration factor is engine-dependent (2.3×–4.0× here).
5. **If full automation is not achievable, say so and quantify the human fraction** rather than reporting an automated number that hides it. Our measured floor is 50%; we report it as a result, not a shortfall.

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9. Reproducibility

Code: github.com/Nasser-Lim/jamo-bench. Data: huggingface.co/datasets/Nasser4963/jamo-gold.

```
git clone https://github.com/Nasser-Lim/jamo-bench && cd jamo-bench
pip install -e ".[dev]"
python scripts/verify_claims.py
```

Every number in this note is recomputed from `release/` (mirrored to the HuggingFace repo above) by this script. If its output disagrees with this document, the document is wrong. The script exists because a data-loading bug in an earlier revision silently dropped 44 human-labelled images and inverted two of this project's conclusions; numbers are therefore never transcribed by hand.

Released artifacts, code layout, and licensing are described in [`RELEASE.md`](RELEASE.md).

Environment note. PaddleOCR 3.x requires `enable_mkldnn=False` on Windows CPU (the default crashes with a PIR/oneDNN error). OpenCV cannot read paths containing Hangul on Windows, so all image loading goes through PIL.

10. Authorship and disclosed tool use

The study design, data collection, annotation, statistical analysis, and every conclusion in this note are the author's own work and are independently checkable against raw data by running `verify_claims.py` (§9) — nothing here is asserted without a script that recomputes it from source. Prose drafting, editing, and the Korean translation were produced with the assistance of a general-purpose AI writing tool, under the author's direction; the author reviewed and takes full responsibility for every claim and figure. No AI system is an author of this work, and no text or figure was generated to imply a result not present in the underlying data.

Two figures (Fig. 1, Fig. 3) contain real outputs from the commercial text-to-image model under study. This is experimental *data*, not editorial content produced for this note — the images are the subject being measured, not an illustration generated to accompany the argument — and this is disclosed in each caption (§1, §3).

Appendix A — Failure composition (human ground truth, n = 300)

Coda class	Correct	Valid but wrong	Not a valid character
no coda (n=90)	70.0%	13.3%	16.7%

simple coda (n=120)	60.8%	15.0%	24.2%
complex coda (n=90)	55.6%	21.1%	23.3%

Valid-syllable production rate by coda class, with bootstrap CI:

Coda class	Rate	95% CI
no coda	83.3%	[75.6, 90.0]
simple coda	75.8%	[67.5, 83.3]
complex coda	76.7%	[67.8, 85.6]

complex − no coda: −6.7pp, CI [−17.8, +4.4] — **not significant**, and not monotonic. We do not claim that complex codas produce more invalid glyphs.

Appendix B — Jamo-position error rates (human ground truth)

Position	Conditional (valid only, n=235)	Unconditional (invalid = all three wrong, n=300)
onset	1.7% [0.4, 3.4]	23.0% [18.3, 28.0]
nucleus	9.8% [6.0, 13.6]	29.3% [24.3, 34.3]
coda	17.0% [12.3, 21.7]	35.0% [29.7, 40.3]

The onset < nucleus < coda ordering holds under both denominators. Note that jamo decomposition is only defined for valid syllables; for invalid glyphs any decomposition is an artefact of whatever the judge snapped to, which is why both denominators are reported.